④ $f(x,y) = -2x^{12}\cos(y) + x$ $\quad x_0 = 10$ $\quad y_0 = \pi/2$

$$L(t) = f(x_0,y_0) + \left.\frac{\partial f}{\partial x}\right|_{x_0,y_0}(x-x_0) + \left.\frac{\partial f}{\partial y}\right|_{x_0,y_0}(y-y_0)$$

$$f(x_0,y_0) = -2(10)^{12}\cos(\pi/2) + 10 = 0 + 10 = 10$$

$$\left.\frac{\partial f}{\partial x}\right|_{x_0,y_0} = \left[-24x^{11}\cos(y) + 1\right]_{x_0,y_0} = -24(10)^{11}\cos(\pi/2) + 1 = 1$$

$$\left.\frac{\partial f}{\partial y}\right|_{x_0,y_0} = \left[2x^{12}\sin(y)\right]_{x_0,y_0} = 2(10)^{12}\sin(\pi/2) = 2\cdot 10^{12}$$

$$L(t) = 10 + (x-10) + (2\cdot 10^{12})(y-\pi/2)$$

$$\boxed{L(t) = x + (2\cdot 10^{12}\cdot y) + \pi\cdot 10^{12}}$$